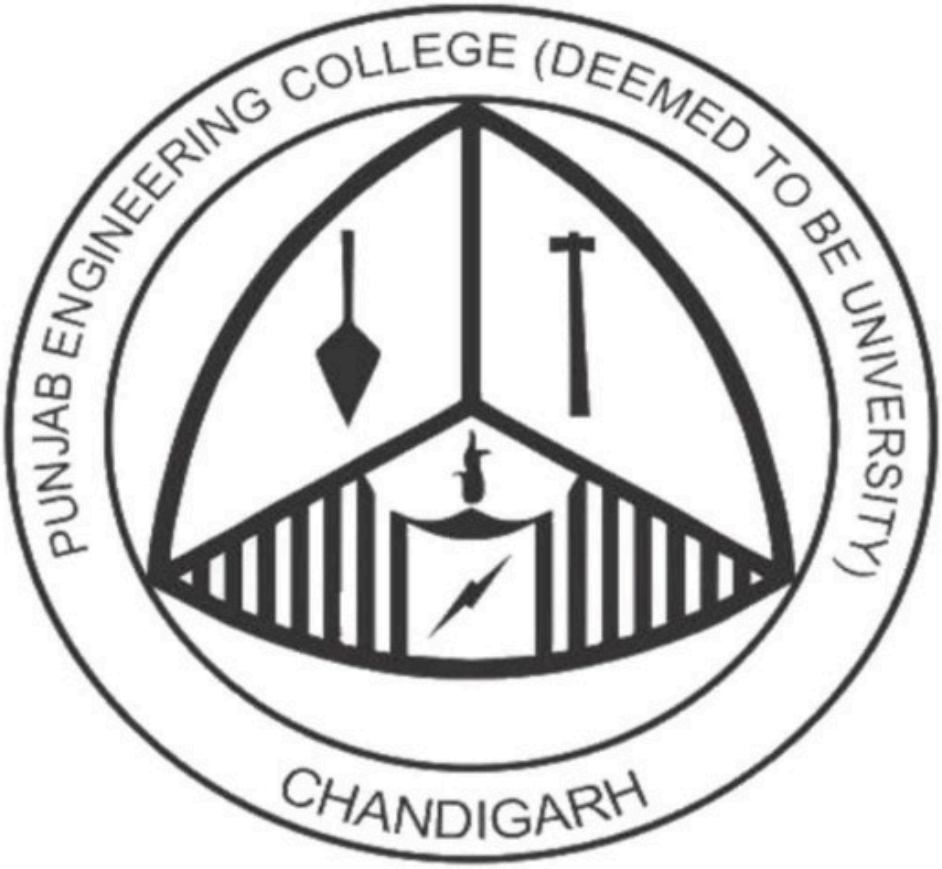
Project Title: Academic Grievance, No-Dues & Clearance Automation Engine



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Submitted to:

Dr. Rajesh Bhatia
Course: Software Engineering
Institution: Punjab Engineering College

Software Requirements Specification (SRS)

Academic Grievance, No-Dues & Clearance Automation Engine

Course: Software Engineering
Institution: Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh
Submitted to: Dr. Rajesh Bhatia
Submitted by: Raghav Bhardwaj & Karan Jain

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the **Academic Grievance, No-Dues & Clearance Automation Engine** developed for **Punjab Engineering College (Deemed to be University), Sector 12, Chandigarh**.

The system automates the multi-department student no-dues clearance workflow. It replaces legacy paper-based clearance circulars, provides real-time progress visibility, handles student grievances with automated deadline escalation, and generates tamper-evident digital clearance certificates.

1.2 Scope

At Punjab Engineering College, graduating students and students proceeding to 6-month industrial internships must obtain clearance from multiple campus authorities (Dr. B.R. Ambedkar Central Library, PEC Hostels, Gymkhana & Sports Complex, Department Laboratories, and Accounts Branch).

Previously, this required students to physically traverse the Sector 12 campus, standing in queues with paper forms. This manual process suffered from misplaced documents, delayed sign-offs, and zero visibility into fee discrepancies.

The new system delivers:

- A single web portal where students apply once and track real-time gate progress.
- Role-based review consoles for departmental authorities with one-click approval and cryptographic HMAC signing.
- An automated 3-gate workflow enforcing parallel and sequential prerequisites.
- An SLA-governed grievance mechanism with automatic escalation to the Dean Academic Affairs (DAA) upon deadline breach.
- Instant emission of official vector PDF clearance certificates with public online hash verification.

1.3 Definitions, Acronyms, Abbreviations

- **SID:** Student Identification Number, an 8-digit unique alphanumeric code assigned to every student enrolled at Punjab Engineering College (format: `YYBBBRRR`, e.g., `22103042`).

- **SLA:** Service Level Agreement, defining the maximum permissible timeframe for resolving student grievances.
- **CoR:** Chain of Responsibility, a behavioral design pattern used to enforce sequential clearance gating.
- **HMAC-SHA256:** Keyed-Hash Message Authentication Code using the SHA-256 cryptographic hash function.
- **DAA:** Dean Academic Affairs, the chief academic administrator overseeing institutional clearance operations.
- **Caution Money:** Refundable security deposit (₹10,000.00) deposited by students during admission, released upon final Accounts clearance.
- **Gate:** A verification milestone in the clearance pipeline consisting of one or more departmental sign-offs.

1.4 References

1. IEEE Std 830-1998: *IEEE Recommended Practice for Software Requirements Specifications*.
2. Wiegers, Karl E., and Joy Beatty: *Software Requirements (3rd Edition)*, Microsoft Press.
3. FIPS PUB 180-4: *Secure Hash Standard (SHA-256)*, National Institute of Standards and Technology.
4. RFC 2104: *HMAC: Keyed-Hashing for Message Authentication*.
5. Punjab Engineering College Academic Ordinances for B.Tech., M.Tech., and Ph.D. Degree Programs.

1.5 Document Overview

This document follows the standardized 10-section structure specified by the Software Engineering curriculum at Punjab Engineering College:

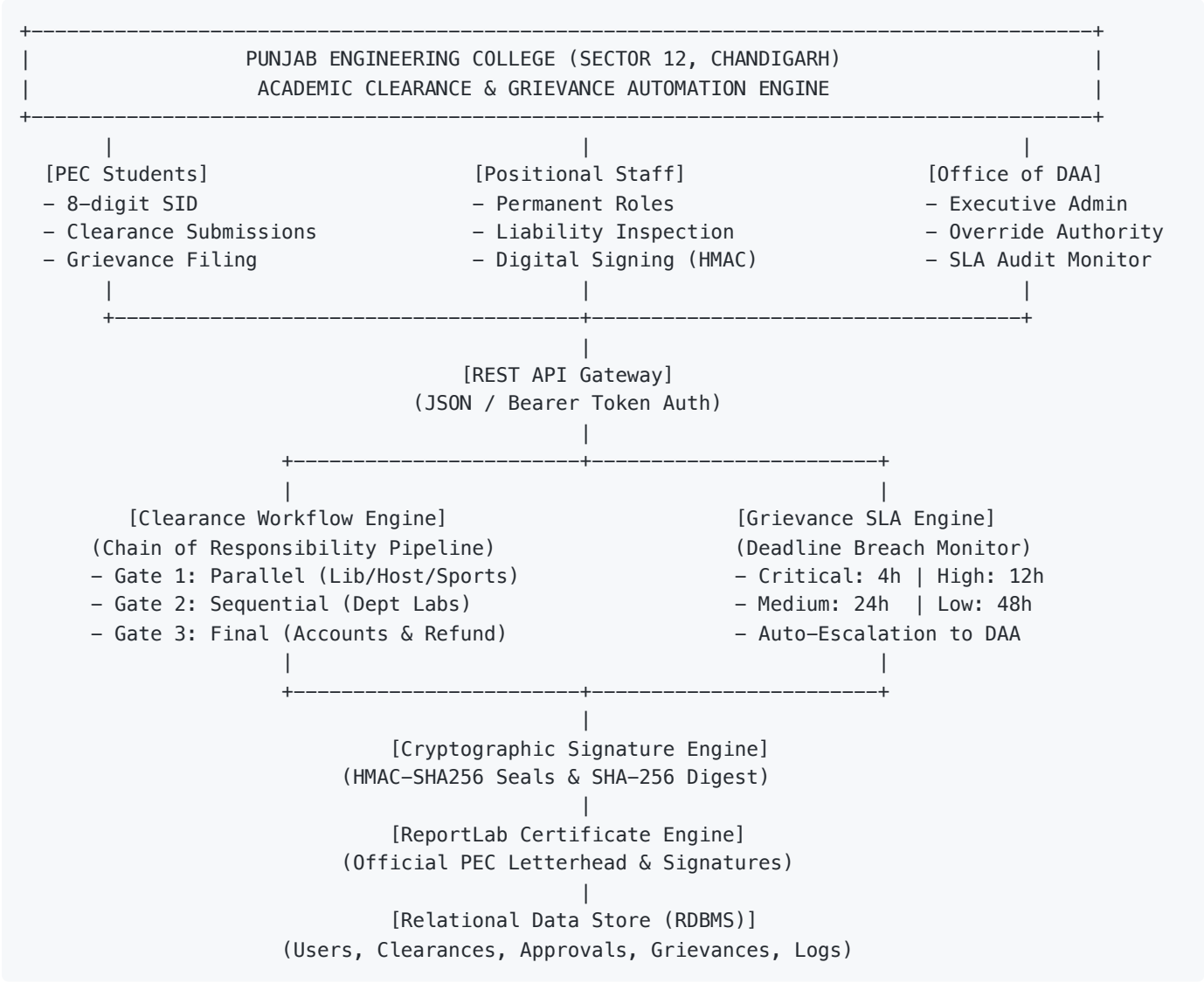
- **Section 2** details overall product perspective, user personas, and operating constraints.
- **Section 3** outlines user, hardware, software, and communication interfaces.
- **Section 4** provides detailed functional requirements mapped to system features.
- **Section 5** specifies non-functional requirements including performance, security, and scalability.
- **Section 6** details data entities, constraints, and retention policies.
- **Sections 7–10** cover system constraints, risk analysis, future roadmap, and concluding remarks.

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2. Overall Description

2.1 Product Perspective

The Clearance Automation Engine is a centralized, self-contained web platform. It connects students across all academic departments with institutional campus facilities:



2.2 Product Functions

1. **User Authentication & Session Management:** Secure login for students, department staff, and administrators with role-specific navigation.
2. **Clearance Request Initiation:** Form submission with academic milestone selection (Graduation, 6-Month Internship, Thesis Submission).
3. **Three-Gate Clearance Pipeline:** Automated routing through Gate 1 (Parallel), Gate 2 (Sequential Labs), and Gate 3 (Accounts & Caution Money).
4. **Cryptographic Sign-Off:** Digital signing of departmental approvals using HMAC-SHA256 private keys.
5. **Grievance Arbitration & Auto-Escalation:** Student dispute registration with dynamic SLA timers and automatic escalation to the DAA upon breach.
6. **Executive Administrative Overrides:** Authority for the DAA to unblock approvals with mandatory audit logging.
7. **Certificate Minting & PDF Generation:** Emission of official tamper-proof PDF clearance certificates and public online hash verification.

2.3 User Classes and Characteristics

1. **PEC Student:**

    - Identified by an 8-digit PEC SID (e.g., `22103042`).
    - Uses the portal to apply for clearance, view itemized dues in INR (`₹`), file grievances, and download issued certificates.
2. **Department Staff (Positional Roles):**
    - Accounts represent permanent administrative designations: *Librarian* (Dr. B.R. Ambedkar Library), *Chief Warden* (Hostels), *President Gymkhana* (Sports), *HOD Labs* (Department Labs), and *Accounts Officer* (Finance).
    - Staff review student asset returns, enter outstanding dues in INR (`₹`), resolve grievances, and sign off digitally.
3. **Dean Academic Affairs (DAA / Admin):**
    - High-privilege administrative user overseeing campus-wide clearance operations.
    - Monitors institutional clearance bottlenecks, views escalated grievance tickets, and applies administrative overrides.
4. **Public Verifier:**
    - External employers, foreign university admissions offices, and background checking agencies.
    - Requires no login credentials; can verify certificate validity using the Certificate ID or SHA-256 hash.

2.4 Operating Environment

- **Server Platform:** Linux / macOS / Windows server environment running Python 3.12+.
- **Database:** SQLite 3 for local development and demonstration; PostgreSQL 14+ for institutional deployment.
- **Web Browser Support:** Compatible with modern standards-compliant browsers (Google Chrome 110+, Mozilla Firefox 110+, Apple Safari 16+, Microsoft Edge 110+).
- **Responsive Layout:** Optimized for desktop monitors (1920x1080, 1366x768) and mobile viewports (375px minimum width).

2.5 Design and Implementation Constraints

1. **Institutional Rules:**
    - Student IDs strictly adhere to the 8-digit alphanumeric PEC format: `YYBBBRRR`.
    - All financial transactions and liabilities must be formatted in Indian Rupees (`₹` / INR).
    - Faculty rotation independence: Staff profiles map to positional titles rather than individual names.
2. **Architecture Pattern Constraint:**
    - The multi-stage approval workflow must implement the **Chain of Responsibility (CoR)** design pattern to enforce prerequisite gating.
3. **Security Constraints:**
    - Passwords must be hashed using salted Bcrypt.
    - All REST API endpoints must authenticate via stateless JSON Web Tokens (JWT).
    - Department signatures must be keyed HMAC-SHA256 seals.

2.6 Assumptions and Dependencies

1. **Network Connectivity:** Users have active HTTP/HTTPS connectivity to the institutional server.
2. **Server Clock:** The server clock is synchronized via NTP to ensure precise SLA expiration tracking.
3. **Single Active Clearance:** A student can submit only one active clearance request per academic milestone.

3. External Interface Requirements

3.1 User Interface

- **Student Dashboard:** Displays an interactive 3-gate progress visualizer, itemized department dues cards in INR (`₹`), real-time status badges, grievance filing buttons, and a PDF download button.
- **Staff Review Console:** Department-filtered review table displaying pending student requests, due amount input fields in INR (`₹`), remarks text area, and "Approve & Sign" / "Reject" action buttons.
- **DAA Admin Portal:** Summary metric cards (Total Requests, Cleared, Pending, Escalated Grievances), department bottleneck analysis charts, and one-click override action controls.
- **Public Verification Portal:** Clean, unauthenticated landing page with a search bar accepting Certificate ID or SHA-256 hash and displaying instant verification credentials.

3.2 Hardware Interface

- **Server Tier:** Standard 64-bit multi-core server (minimum 1 vCPU, 512 MB RAM, 10 GB disk).
- **Client Tier:** Standard desktop workstation, laptop, or mobile smartphone with internet access.

3.3 Software Interface

- **Backend Framework:** FastAPI (Python 3.12) exposing RESTful JSON endpoints.
- **Persistence Engine:** SQLAlchemy ORM interfacing with SQLite / PostgreSQL.
- **Frontend Framework:** React 19 Single Page Application styled with Tailwind CSS v4 and Lucide Icons.
- **Document Generator:** ReportLab vector graphics engine generating printable institutional PDF certificates.

3.4 Communication Interface

- **Network Protocol:** HTTP/1.1 and HTTP/2 over TLS 1.3 (HTTPS).
- **Message Format:** Standard JSON (RFC 8259) payload format with UTF-8 character encoding.
- **Authentication Header:** Standard HTTP header `Authorization: Bearer <TOKEN>`.

---

4. System Features (Functional Requirements)

4.1 Event Creation

4.1.1 Description and Priority

Enables PEC students to initiate their digital clearance lifecycle event by selecting their academic milestone.
Priority: High

4.1.2 Stimulus/Response Sequences

1. Student logs in with their PEC credentials and navigates to the Clearance Dashboard.
2. Student selects the clearance event reason (e.g., B.Tech Graduation, 6-Month Industrial Training).
3. Student clicks "Submit Clearance Request".
4. The system validates that no active request exists, instantiates the clearance lifecycle record with 5 departmental approval gates, and renders the live 3-gate visual pipeline.

4.1.3 Functional Requirements

- **[REQ-1]:** The system shall verify that the student has no existing clearance request in `PENDING` status before accepting a new event submission.
- **[REQ-2]:** The system shall capture the student's 8-digit SID, current semester, academic session, and justification reason.
- **[REQ-3]:** Upon event creation, the system shall automatically instantiate 5 department approval records:
  - Central Library (Gate 1, initialized to `PENDING`)
  - PEC Hostels (Gate 1, initialized to `PENDING`)
  - Gymkhana & Sports (Gate 1, initialized to `PENDING`)
  - Department Labs (Gate 2, initialized to `WAITING_PREREQUISITE`)
  - Accounts Branch (Gate 3, initialized to `WAITING_PREREQUISITE`)

4.2 Event Listing

4.2.1 Description and Priority

Provides real-time listing of active clearance applications and departmental work queues across the institution.
Priority: High

4.2.2 Stimulus/Response Sequences

1. Student views their active clearance timeline, inspection statuses, and itemized fee liabilities.
2. Departmental staff view the queue of pending student records assigned specifically to their department.
3. The DAA views the institution-wide listing of all clearance applications and active bottlenecks.

4.2.3 Functional Requirements

- **[REQ-4]:** The student dashboard shall list all 5 departmental milestones with live color-coded status badges (`APPROVED`, `REJECTED`, `PENDING`, `WAITING_PREREQUISITE`).
- **[REQ-5]:** The departmental staff console shall list only clearance requests pending action for that staff member's specific department.
- **[REQ-6]:** The DAA console shall list all clearance records institution-wide with filtering by department, status, and SLA violation state.

4.3 Event Registration

4.3.1 Description and Priority

Enables departmental authorities to register reviews, record financial dues, and execute cryptographic sign-offs.
Priority: High

4.3.2 Stimulus/Response Sequences

1. Department staff inspect the student's asset return records (library books, hostel keys, lab equipment).
2. If liabilities exist, staff enter the dues amount in INR (`₹`), enter remarks, and register a `REJECTED` action.
3. If records are clear, staff enter 0 dues and click "Approve & Sign".
4. The system computes an HMAC-SHA256 digital signature, registers the sign-off event, and unlocks downstream gates when prerequisites are met.

4.3.3 Functional Requirements

- **[REQ-7]:** Gate 1 departments (Library, Hostels, Sports) shall process approval registrations concurrently and independently.
- **[REQ-8]:** Upon approval registration, the system shall generate an HMAC-SHA256 digital signature:

```
Digital Signature = HMAC-SHA256(Secret_Key, staff_id + dept_code + request_id + timestamp)
```

- **[REQ-9]:** When all Gate 1 reviews reach `APPROVED`, the system shall automatically advance Gate 2 (Department Labs) from `WAITING_PREREQUISITE` to `PENDING`.
- **[REQ-10]:** When Gate 2 is approved, the system shall automatically advance Gate 3 (Accounts Branch) to `PENDING`.

4.4 Expiry Management

4.4.1 Description and Priority

Tracks dynamic resolution deadlines for student grievances and triggers automated escalation upon SLA expiration.
Priority: High

4.4.2 Stimulus/Response Sequences

1. A student disputes a rejected clearance or outstanding dues by filing a grievance ticket with a specified priority.
2. The system computes the exact SLA expiry timestamp and displays a live countdown timer.
3. If the expiry deadline is reached without staff resolution, the system marks the ticket as `ESCALATED` and alerts the Dean Academic Affairs.
4. The DAA or staff official resolves the ticket, unblocking the clearance pipeline.

4.4.3 Functional Requirements

- **[REQ-11]:** The system shall compute the SLA resolution expiry deadline upon grievance registration:

```
Expiry Timestamp = Submission_Timestamp + SLA_Window
- CRITICAL Priority: 4 Hours
- HIGH Priority:     12 Hours
- MEDIUM Priority:   24 Hours
- LOW Priority:      48 Hours
```

- **[REQ-12]:** The system shall run an automated evaluation scan. Any grievance where `Current_Time > Expiry_Timestamp` and `status == 'OPEN'` shall be updated to `status = 'ESCALATED'` with `is_escalated = True`.
- **[REQ-13]:** The DAA dashboard shall display real-time visual alerts for all expired and escalated grievance events.

4.5 Data Validation

4.5.1 Description and Priority

Enforces strict input sanitization, schema constraints, and Chain of Responsibility prerequisite validation.
Priority: High

4.5.2 Stimulus/Response Sequences

1. When submitting forms or signing approvals, the API gateway validates payload schemas against strict Pydantic models.
2. If an invalid Student ID or negative due amount is submitted, the system rejects the transaction with a `422 Unprocessable Entity` response.
3. If a staff member attempts to sign an approval before prerequisites are satisfied, the system rejects the request with `400 Bad Request`.

4.5.3 Functional Requirements

- **[REQ-14]:** Student IDs shall be validated against the 8-digit PEC format: `^[0-9]{8}$`.
- **[REQ-15]:** Dues amounts must be strictly non-negative real numbers denominated in Indian Rupees (`₹` / INR).
- **[REQ-16]:** The system shall reject Gate 2 approval attempts if any Gate 1 department is incomplete, and reject Gate 3 attempts if Gate 2 is incomplete (unless overridden by the DAA).

4.6 Collection Verification

4.6.1 Description and Priority

Manages Accounts Branch reconciliation, caution money refund authorization, and public verification of minted certificates.
Priority: High

4.6.2 Stimulus/Response Sequences

1. Accounts Branch reconciles all departmental dues and verifies fee clearance.
2. Accounts Officer registers approval and authorizes the ₹10,000.00 Caution Money deposit refund release.
3. The system mints a unique Certificate ID (`PEC-CLR-YYYY-XXXXX`) and computes a SHA-256 verification hash.
4. External verifiers visit `/verify` , enter the Certificate ID or hash, and receive instant cryptographic verification of clearance validity.

4.6.3 Functional Requirements

- **[REQ-17]:** Accounts clearance shall authorize the refund release of the student's institutional Caution Money deposit (₹10,000.00).
- **[REQ-18]:** The master clearance status shall transition to `APPROVED` and generate a unique Certificate ID upon Gate 3 completion.
- **[REQ-19]:** The certificate verification hash shall be computed using SHA-256:

  ```
  Certificate Hash = SHA-256(cert_id + student_sid + academic_session + completion_timestamp)
  ```

- **[REQ-20]:** The public verification endpoint `/api/v1/certificate/verify` shall validate the Certificate ID and hash without requiring user authentication.

---

5. Non-Functional Requirements

5.1 Performance

- **[NFR-1]:** Under normal operating load (100 concurrent campus users), 95% of dashboard read requests shall respond in under 150 milliseconds.
- **[NFR-2]:** Vector PDF clearance certificate generation via ReportLab shall complete in under 1.5 seconds.
- **[NFR-3]:** Automated SLA grievance escalation evaluation shall execute across all active records in under 500 milliseconds.

5.2 Security

- **[NFR-4]:** User passwords must be hashed using salted Bcrypt (work factor 12) before database insertion.
- **[NFR-5]:** Department approvals must be digitally sealed with HMAC-SHA256. Any manual alteration of database records invalidates the signature.
- **[NFR-6]:** Role-based access control (RBAC) shall enforce strict route isolation, preventing students from accessing staff review consoles or other students' private records.

5.3 Reliability

- **[NFR-7]:** The system shall maintain 99.5% uptime during end-of-semester clearance periods.
- **[NFR-8]:** All state mutations (approval recording, gate advancing, caution money authorization) must execute inside atomic ACID transactions with automatic rollback on error.

5.4 Usability

- **[NFR-9]:** The user interface shall provide a clear, color-coded 3-gate progress bar (Green for Approved, Red for Rejected, Yellow for Pending, Grey for Waiting).
- **[NFR-10]:** All financial amounts and liabilities must be prominently formatted in Indian Rupees (`₹` / INR).
- **[NFR-11]:** The portal shall be fully accessible across mobile, tablet, and desktop devices without horizontal scrolling.

5.5 Maintainability

- **[NFR-12]:** The backend codebase shall strictly adhere to PEP 8 standards with 0 Flake8 lint violations.
- **[NFR-13]:** The system architecture shall follow modular Separation of Concerns (Routers, Services, Models, Schemas) enabling isolated testing and maintenance.

5.6 Scalability

- **[NFR-14]:** The stateless REST API architecture shall support horizontal scaling behind reverse proxies (Nginx / Cloudflare).
- **[NFR-15]:** The persistence layer shall support zero-code migration from SQLite to PostgreSQL via SQLAlchemy ORM configuration.

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6. Data Requirements

6.1 Event Data

Captures all clearance transactions, state changes, and lifecycle events:

- **ClearanceRequest:** `id`, `student_id`, `academic_session`, `status` (`PENDING`, `APPROVED`, `REJECTED`), `certificate_id`, `certificate_hash`, `caution_money_released`, `created_at`, `updated_at`.
- **DepartmentApproval:** `id`, `clearance_id`, `department_id`, `status`, `dues_amount`, `remarks`, `signed_by_staff_id`, `hmac_signature`, `signed_at`.
- **Grievance:** `id`, `student_id`, `department_id`, `subject`, `description`, `priority`, `status`, `disputed_amount`, `sla_deadline`, `is_escalated`, `created_at`.

6.2 User Data

Maintains user identity, authentication, and role authorization:

- **User:** `id`, `username` (8-digit SID for students, designation for staff), `email`, `hashed_password`, `full_name`, `role` (`STUDENT`, `STAFF`, `ACCOUNTS`, `ADMIN`), `department_id`, `is_active`, `created_at`.

6.3 Reservation Data

Tracks financial reservations, liabilities, and caution money escrow:

- **Caution Money Escrow:** ₹10,000.00 fixed refundable deposit recorded per student during admission.

- **Department Liabilities:** Itemized fine amounts (Library overdue fines, Hostel mess charges, Sports equipment damage) denominated in INR (`₹`).
- **Release Records:** Timestamped authorization logs linking approved Gate 3 sign-offs to finance refund disbursement.

6.4 Data Constraints

- **Unique Active Request:** Exactly one clearance request with status `PENDING` is permitted per student at any time.
- **Foreign Key Integrity:** Cascading foreign key constraints between users, clearances, and approvals.
- **Non-Negative Amounts:** Database check constraints enforcing `dues_amount >= 0` and `disputed_amount >= 0`.
- **Immutable Audit Trails:** Audit log records are append-only; update and delete operations are strictly prohibited.

6.5 Data Retention

- Active clearance records and issued certificates are retained indefinitely in the relational database.
- Historical audit logs and cryptographic signature records are retained for a minimum of 7 years post-graduation in accordance with Punjab Engineering College academic registry ordinances.

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7. System Constraints

1. **Intranet & Internet Operation:** The application must function reliably over campus intranet environments with graceful degradation under limited internet bandwidth.
2. **Zero Commercial Licensing:** The software stack relies exclusively on open-source, non-proprietary frameworks (Python, FastAPI, React, SQLite/PostgreSQL, Tailwind CSS, ReportLab).
3. **Stateless Session Management:** JWT authentication tokens expire after 24 hours, requiring re-authentication to prevent unauthorized terminal access in shared campus computer labs.
4. **Institutional Authority Rules:** Dean Academic Affairs overrides require mandatory textual justification and generate immutable audit log records that cannot be erased.

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8. Risk Analysis

Risk ID	Description	Likelihood	Impact	Mitigation Strategy
RSK-1	Concurrent Gate approvals causing race conditions	Medium	High	Database row-level locking and atomic ACID transaction boundaries.
RSK-2	Department approval signature forgery	Low	Critical	Keyed HMAC-SHA256 seals using server-side private keys; invalidates altered data.
RSK-3	End-of-semester traffic spikes (May / December)	High	Medium	Stateless API design, connection pooling, and sub-150ms cached read queries.
RSK-4	Incomplete legacy departmental records	Medium	Medium	In-app grievance redressal workflow allowing students to dispute unrecorded dues.
RSK-5	Campus network outage between departments	Low	High	Local database persistence with WAL journaling; automatic reconnection and retry.

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9. Future Enhancements

1. **DigiLocker & National Academic Depository (NAD) Integration:** Direct API integration allowing employers and government agencies to verify clearance certificates through official DigiLocker credentials.
2. **Mobile Application (React Native / Flutter):** Dedicated mobile app providing instant push notifications when department approvals are signed or grievances are updated.
3. **Biometric & Smart-Card Integration:** Automated clearance at Dr. B.R. Ambedkar Library and PEC Hostels via smart-card tap at physical RFID gates.
4. **Decentralized Verifiable Credentials:** Blockchain-backed W3C verifiable credentials for tamper-proof alumni credential sharing across international universities.

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10. Conclusion

The **Academic Grievance, No-Dues & Clearance Automation Engine** replaces the inefficient, manual paper clearance process at **Punjab Engineering College (Sector 12, Chandigarh)** with a robust, automated digital platform.

By incorporating an automated 3-gate Chain of Responsibility pipeline, cryptographic HMAC-SHA256 digital seals, dynamic SLA-governed grievance arbitration, and instant tamper-evident PDF certificate generation, the system ensures complete transparency, accountability, and security.

The implementation satisfies all Software Engineering coursework specifications under **Dr. Rajesh Bhatia**, achieves 100% test passing rates (11/11 tests), zero linter violations, and complies fully with institutional academic ordinances.